MILITARY SPECIFICATION SHEET

INTERCOMMUNICATIONS SETS, AN/AIC-14 and AN/AIC-14A

MONITOR AMPLIFIER MODULE

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope. This specification sheet provides performance and physical characteristics for the Monitor Amplifier Module used in station Interphone Control panel C-4951/AIC-14A.

1.2 Military Part Number. The military part number shall be M22352/9-1

2. APPLICABLE DOCUMENTS

Refer to 2.2 of MIL-I-22352(AS) for a listing of applicable documents. The requirements specified herein shall have precedence over all referenced documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The Monitor Amplifier Module shall provide amplification to two independent audio signals with gain characteristics as specified herein when used in Station Interphone Control Panel or C-4951/AIC-14A.

3.1.1 Item Diagram. The Monitor Amplifier Module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface Definition. The Monitor Amplifier Module input and output signals will normally display the characteristics presented in Table I.

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TABLE I. Signal characteristics

Signal Name	Characteristics
Inputs:	
Monitor Relay (RELAY)	Relay logic: 1) high impedance open 2) low impedance to ground
Enable (ENABLE)	Relay logic: 1) open 2) short to ground
Monitor (MONITOR)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Selected (SELECTED)	Voice range audio (300 to 7000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Power (PWR)	27.5 ±0.5 Vdc
Outputs:	
Enable (ENABLE)	Binary logic: Hi) above 26 Vdc Lo) less than 2 Vdc
Monitor (MONITOR)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 10 Vac.
Selected (SELECTED)	Voice and sidetone range audio (300 to 7000 Hz) signal ranging from 0 to 0.45 Vac.
Grounds:	
Power (PWR GND)	Power return. See 3.4.4.2 of MIL-I-22352(AS).
System (SYS GND)	Signal return. See 3.4.4.3 of MIL-I-22352(AS).

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3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 Relay (RELAY). The Monitor Amplifier Module shall have a RELAY input for activation of a monitor relay. The monitor relay shall activate when a ground or relay logic low is applied to the RELAY input.

3.2.1.1.2 Enable (ENABLE). The Monitor Amplifier Module shall have an ENABLE input for logic switching. The enable gate shall be enabled when an input above 0.2 Vdc is applied.

3.2.1.1.3 Monitor (MONITOR). The Monitor Amplifier Module shall have a MONITOR audio input circuit having the following input characteristics:

3.2.1.1.3.1 Signal Frequency. The MONITOR input shall accept input drive signals which range from 300 to 7,000 Hz.

3.2.1.1.3.2 Signal Amplitude. The MONITOR input shall accept input drive signals which range in amplitude from 0.01 to 0.10 Vac. The MONITOR input shall be capable of withstanding overvoltage inputs of up to 1.0 Vac without sustaining damage. The MONITOR input shall be capable of operating within specification when the signal input contains a dc component of up to 2 Volts.

3.2.1.1.3.3 Input Impedance. The MONITOR input shall be designed to produce the specified outputs when driven by a signal source having an output impedance of 10,000 ohms.

3.2.1.1.4 Selected (SELECTED). The SELECTED input characteristics shall be as described in Section 3.2.1.1.3 for the monitor input. The SELECTED signal shall be applied to the Select amplifier.

3.2.1.1.5 Power (PWR). The Monitor Amplifier Module shall accept and operate within the constraints of this specification when supplied with +27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1. The maximum load presented by the Monitor Amplifier Module shall not exceed 300 milliamperes.

3.2.1.2 Output Characteristics. The Monitor Amplifier Module shall have the following output characteristics:

3.2.1.2.1 Enable (ENABLE). The Monitor Amplifier Module shall have an ENABLE output which is the binary complement of the input. The output amplitude shall be:

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State	Voltage
HI	Greater than 26 Vdc
LO	Less than 2 Vdc

3.2.1.2.2 Monitor (MONITOR). The MONITOR output shall have two output conditions. In condition 1, the monitor relay is energized and a normal output as described below shall be provided. In condition 2, the monitor relay is de-energized and the output is disabled, presenting an open circuit. When open circuit, the input/output isolation shall be at least 40 dB.

The MONITOR output shall have the following characteristics when the monitor relay is energized:

3.2.1.2.2.1 Frequency and Amplitude. The monitor signal output shall exceed the input amplitude at 1,000 Hz by 40 ±3 dB when terminated in a 1,200 ohm resistive load. Within the frequency range from 300 to 7,000 Hz the signal amplitude shall not vary more than 2 dB above or below the value at 1,000 Hz.

3.2.1.2.2.2 Output Impedance. The MONITOR output shall have an impedance of 1200 ohms ±10 percent.

3.2.1.2.2.3 Distortion. The total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed 5 percent under standard conditions as defined in Section 3.1a of MIL-STD-810. The total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed 10 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment.

3.2.1.2.2.4 Amplifier Stability. The amplifier which produces the MONITOR output signal shall be stable and shall show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, and/or any combination thereof. No damage shall result from amplifier operation with inputs and/or outputs terminated in either open or short circuits.

3.2.1.2.2.4 Internal Noise. The Monitor amplifier shall have an output of less than 5 millivolts when operated throughout the environmental envelope of MIL-E-5400 for Class 2 equipment with the MONITOR input circuit shorted.

3.2.1.2.2.6 Susceptibility to Stray Magnetic Fields. The Monitor amplifier shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz.

3.2.1.2.3 Selected (SELECTED). The SELECTED output shall have the following characteristics:

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3.2.1.2.3.1 Frequency and Amplitude. The SELECTED signal output shall exceed the input amplitude at 1,000 Hz by 13 ±2dB when terminated in a 1,200 ohm resistive load. Within the frequency range from 300 to 7,000 Hz, the signal amplitude shall not vary more than 2dB above or below the value at 1,000 Hz.

3.2.1.2.3.2 Output Impedance. The selected output shall have an impedance of 1200 ohms ±10 percent.

3.2.1.2.3.3 Distortion. The total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed 5 percent under standard conditions as defined in 3.1a of MIL-STD-810. The total harmonic distortion over the frequency range of 300 to 7,000 Hz shall not exceed 10 percent throughout the environmental envelope of MIL-E-5400 for Class 2 equipment.

3.2.1.2.3.4 Amplifier Stability. The amplifier which produces the SELECTED signal shall be stable and shall show no evidence of regeneration or oscillation when operated with input and output circuits terminated in the specified impedance, open circuit, short circuit, and/or combination thereof. No damage shall result from amplifier operation with inputs and/or outputs terminated in either open or short circuits.

3.2.1.2.3.4 Internal Noise. The amplifier shall have an output of less than 5 millivolts when operated throughout the environmental envelope of MIL-E-5400 for Class 2 equipment with the input circuit shorted.

3.2.1.2.3.6 Susceptibility to Stray Magnetic Fields. The amplifier shall be shielded or so designed that when oriented in any position within a magnetic field of 2 Gauss in the frequency range of 400 to 2,800 Hz, the output from the amplifier shall not be greater than 5 millivolts at 400 Hz. Between 400 and 2,800 Hz, the output shall not exceed a linear increase up to 50 millivolts at 2,800 Hz.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The Monitor Amplifier Module shall consist of a single encapsulated module having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The Monitor Amplifier Module shall weigh no more than 3.0 ounces.

3.3 Design and Construction. The design and construction of the Monitor Amplifier Module shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

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4. QUALITY ASSURANCE PROVISIONS
See MIL-I-22352(AS).

4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

4.2 Test Equipment. See 4.3.3.1 of MIL-I-22352(AS).

4.3 Test Methods.

4.3.1 Examination of product. See 4.3.3.2.1 of MIL-I-22352(AS).

4.3.2 Operational test. See 4.3.3.2.2 of MIL-I-22352(AS).

4.3.2.1 Input Impedance. Verify that the input impedance in accordance with 3.2.1.1.3.3 herein.

4.3.2.2 Gain. Verify that the amplifier gain at 1000 Hz is in accordance with 3.2.1.2.2.1 herein.

4.3.2.3 Frequency Response. Holding the input signal constant vary the frequency between 300 Hz and 7000 Hz. Verify that the amplifier gain meets the requirements of 3.2.1.2.2.1 herein.

4.3.2.4 Linearity. Using a 1000 Hz input signal which is varied between 0.01 Vac and 0.10 Vac, verify that the amplifier meets the requirements of 3.2.1.1.3.2.

4.3.2.5 Distortion. Verify the requirements of 3.2.1.2.2.3 herein.

4.3.2.6 Amplifier stability. Verify the requirements of 3.2.1.2.2.4 herein.

4.3.2.7 Internal noise. Verify the requirements of 3.2.1.2.2.5 herein.

4.3.2.8 Susceptibility to stray magnetic field. Verify the performance of 3.2.1.2.2.6 herein.

Preparing Activity:
Navy - AS

Project No. 5831-N042-9

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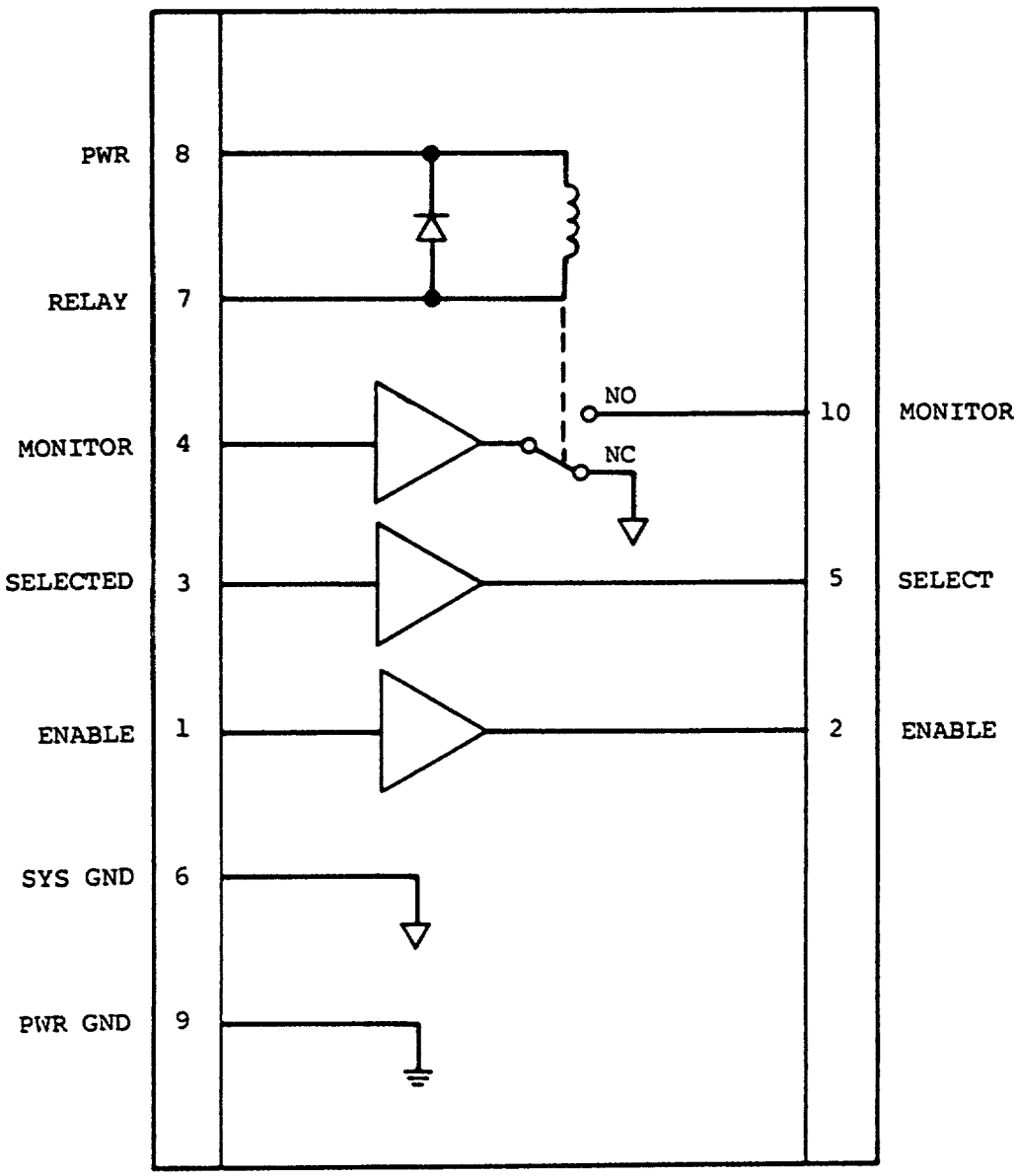
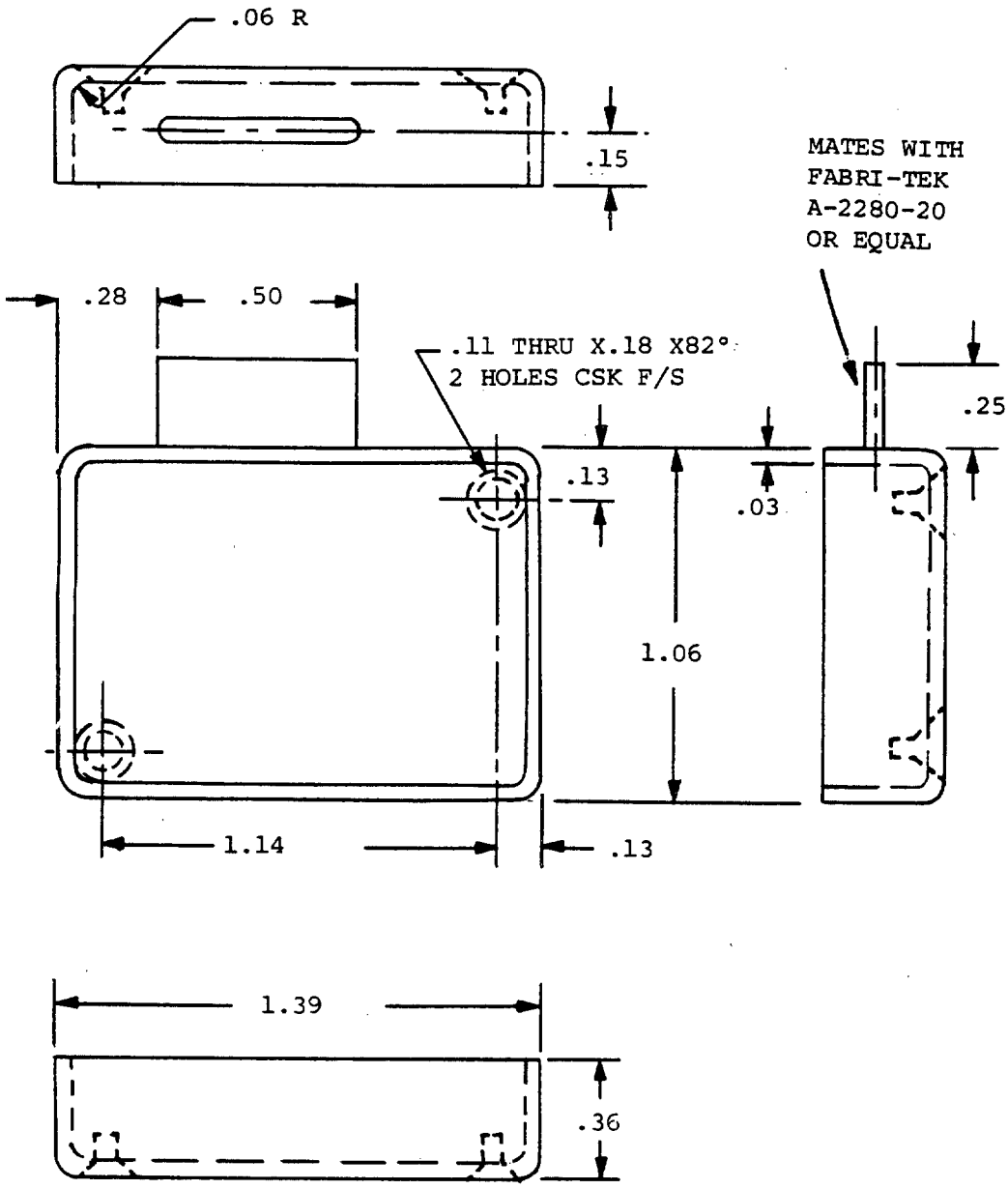


FIGURE 1. Monitor amplifier module functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. Monitor amplifier module outline drawing.

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